Algorithmic Geometry Fall, 2026

TD 2

1. Let $(X, \mathcal{R})$ be a set system, with $n = |X|$ and $m = |\mathcal{R}|$, and $\epsilon > 0$ a parameter. Each $x_i \in X$ has a weight $\omega_i \geq 0$, and for any $S \subseteq X$, set $\Omega(S) = \sum_{x_i \in S} \omega_i$. (Assume $m \geq 2$ and $\Omega(X) > 0$.)

   Construct a random sample $R \subseteq X$ by picking each $x_i \in X$ independently with probability

$$\min\left\{1, \frac{5 \ln m}{\epsilon} \frac{\omega_i}{\Omega(X)}\right\}.$$

   Prove that with probability at least $1 - 1/m^4$, $R \cap S \neq \emptyset$ for each $S \in \mathcal{R}$ with $\Omega(S) \geq \epsilon\, \Omega(X)$.

Solution.

Fix a $S \in \mathcal{R}$ with $\Omega(S) \geq \epsilon \Omega(X)$. If $\frac{5 \ln m}{\epsilon} \frac{\omega_i}{\Omega(X)} \geq 1$ for an $x_i \in S$, then x_i is picked in R and we're done. Otherwise, we have

$$\Pr[R \text{ does not contain any element of } S] = \prod_{x_i \in S} \left(1 - \frac{5 \ln m}{\epsilon} \frac{\omega_i}{\Omega(X)}\right)$$

$$\leq \prod_{x_i \in S} \exp\left(-\frac{5 \ln m}{\epsilon} \frac{\omega_i}{\Omega(X)}\right) = \exp\left(-\frac{5 \ln m}{\epsilon} \cdot \sum_{x_i \in S} \frac{\omega_i}{\Omega(X)}\right) \leq \exp\left(-\frac{5 \ln m}{\epsilon} \cdot \epsilon\right) = \frac{1}{m^5}.$$

By the union bound over all $|\mathcal{R}|$ sets, we conclude that the probability of failure is at most $1/m^4$.

2. Consider the following setup for the experts problem.

   Let $[n]$ be a set of n experts. At each iteration $t = 1, \ldots, m$, each expert makes a prediction for the t-th iteration. After the expert predictions, the loss of each expert is revealed. That is, there is a loss function

   $$\ell^t : \{1, \ldots, n\} \to \{0, 1\}$$

   that assigns a 0-1 loss to the prediction of every expert in the t-th iteration.

   Consider the following randomized algorithm to make predictions that minimize our total loss.

   At the start of iteration t, let $P_t \subseteq [n]$ be the set of experts with the minimum loss so far.

   Pick an expert uniformly at random from P_t, and follow its prediction.

   Prove that

   $$\mathrm{E}\left[\text{our total loss}\right] = O\left(1 + \log n\right) \cdot (L + 1),$$

   where L is the total loss of the best expert.

Solution. The key is to realize that the set P_t changes in only two very specific ways at each iteration.

Case 1: some expert in P_t gets 0 loss at iteration t. Then it must be that $P_{t+1} \subseteq P_t$. As the loss ℓ_i^t is $\{0, 1\}$, those experts in $[n] \setminus P_t$ cannot 'catch up' to the experts in P_t who made the right prediction (with 0 loss) at iteration t.

Case 2: all experts in P_t have loss 1 at iteration t. Then it must be that $P_t \subseteq P_{t+1}$. Some experts in $[n] \setminus P_t$ may catch up to the experts in P_t.

In this case the total loss of the best expert at the end of iteration t is one plus the total loss of the best expert at the end of iteration $t - 1$.

The total loss of the best expert at the end is the number of times **Case 2** occurred. Say **Case 2** occurs L times.

On the other hand, our algorithm definitely has a loss of 1 for each occurrence of **Case 2**. Now we show that our expected loss from all **Case 1** iterations between two consecutive **Case 2** iterations is $O(\log n)$.

Let's say from iterations t_1 to t_2, **Case 1** occurs. Then we have

$$\begin{aligned}
\mathrm{E}\left[\text{our total loss in iterations } [t_1, t_2]\right] &= \sum_{t=t_1}^{t_2} \frac{|P_t| - |P_{t+1}|}{|P_t|} = \sum_{t=t_1}^{t_2} \sum_{k=|P_{t+1}|+1}^{|P_t|} \frac{1}{|P_t|} \\
&\leq \sum_{t=t_1}^{t_2} \sum_{k=|P_{t+1}|+1}^{|P_t|} \frac{1}{k} = \sum_{k=|P_{t_2}|+1}^{|P_{t_1}|} \frac{1}{k} = O\left(\log n\right).
\end{aligned}$$

Thus our total expected loss is L plus the $O\left(\log n\right)$ loss for each of the $L + 1$ 'blocks' of **Case 1**'s.